Pet Detective

Pet Detective tracks your pet's movement in and out of a pet door. The time your pet goes in or out is displayed on the monitor and time stamped in a log. The information appears on the unit's 1.9" LCD monitor.

WiFi is used to get the time and maintain time accurately. Using WiFi, Pet Detective can also display your pet's movement on a browser running on your local network.

Features include

- Location of pet, inside or outside, with time of occurrence
- Percent of time spent outside vs. inside
- View of instant sensor status
- Log of last 32 pet movement directions with a date/time stamp
- WiFi Status and WiFi Reset option
- Automatic time set, even after power failure or DST change
- View the status and log via browser on your local network
- Sensor calibration and distance monitor
- 12 or 24-hour mode selection
- Time Zone setting for 7 North America zones and DST options
  (a code change is required for other time zones)
- Choice of colors for Inside and Outside status: Green, Blue or Amber

Skills Needed

This is a rather complex project, but we've done all the hard work in design! You will need to be able to print 3D parts; make or buy the cover lens; solder a few wires; perform simple mechanical assembly; install the software; mount the unit above your pet door; and enjoy!

Parts List

These are the project parts. You may have some already in your collection or find other sources for the parts. Substituting alternative parts may require changing the case design or software. Links are for sources we use, but we have no affiliation with any of these vendors. We also made a partial kit of parts to save time and money. Prices were as of March 2026.

Electrical Parts

- ESP32 Dev Module, includes 1.9" LCD display ($16)
  https://www.amazon.com/dp/B0D6QXC813

- VL53L5X V2 8x8 Wide Angle Time-of-Flight Laser sensor module ($17-19)
  https://www.walmart.com/ip/8565454261 **OR**
  https://www.amazon.com/QERPON-Multiple-VL53L5CX-Distance-Measuring/dp/B0F1TFWVNL
-

- 180 degree USB C adapter https://www.amazon.com/dp/B0BRFZHJVW ($7 for 2-pack, optional)

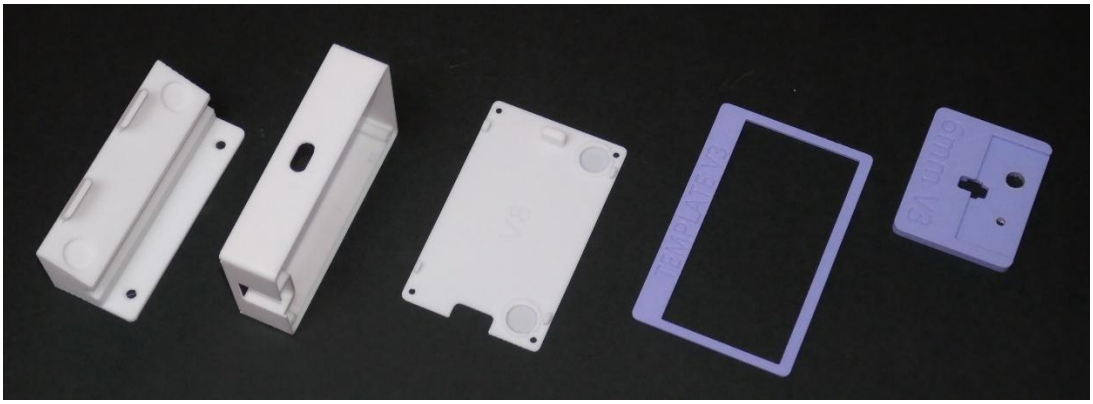
- USB C cable (any cable type will work)
- USB power adapter (It requires less than 10 mA, so any adapter will work)
  OR
  USB Power bank battery – This unit requires so little power, a 7500 mA power bank should power Pet Detective for a month before you'd need to recharge it.

Other Parts

- Hard plastic button (optional, see text, included in parts kit)
- 4 – 10mm diameter x 1mm magnets https://www.amazon.com/dp/B0CZ94J8BS ($8 for 100 magnets)
- 6 – M2.3 x 4mm Flat head self tapping screws (M2.5 x 4 mm ok as well)
- 2 –M3 x 8 mm flat head wood screws (for mounting bracket)
- 2 - M4 x 50 mm round-head wood screws (for optional extended mount to door)
- 1 - Male/Female connector pair with wires (optional, $8 for 10 pairs) https://www.amazon.com/dp/B0DNTK1S9L
  OR
  4 - 3" (8 mm) 26-28 gauge wires, ideally in 4 colors

3D Printed Parts

- Case angle door mount
- Case front
- Case back
- Cover lens template (not needed for laser cut lens)
- Cover lens hole template



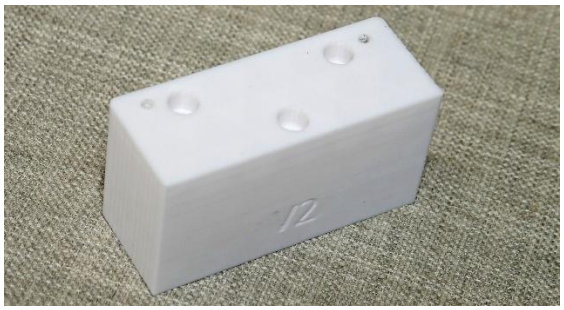
Door mount, case front, case back, and cover lens template and hole template

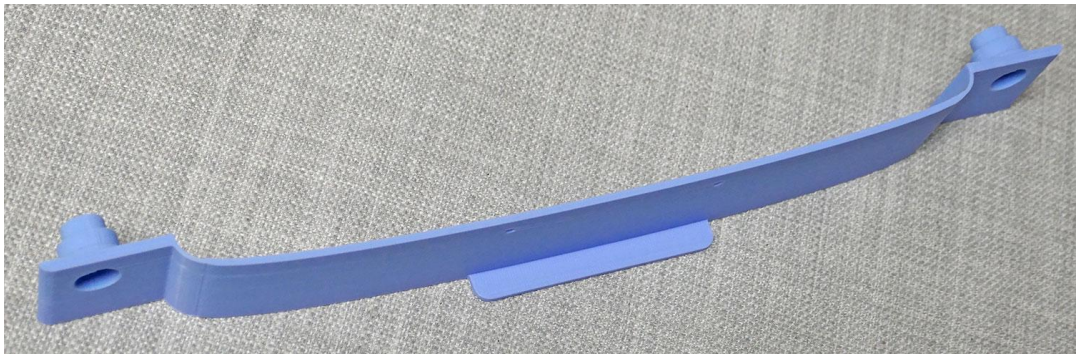
- Blackout LCD mask

Blackout LCD Mask front facing

- Button holder (for hard button, not shown)

Optional 3D Printed Parts

- Extended mount for thicker pet doors such as OnlyCat Microchip, PetSafe SmartDoor, Sure Microchip Connect, etc.
- 
- High Tech medium size pet door mount and covers (covers not shown)



High Tech pet door mount

Hand cut or Laser Cut Cover Lens

- 1/16” cast acrylic Smoke Grey #2074, 6” x 12” sheet, ($5 + $21 shipping) https://www.estreetplastics.com/Gray-Smoked-Plexiglass-Acrylic-2074-1-16-p/x-t81060.htm

  OR

  1/16” Polycarbonate Transparent Grey, 12” x 12” sheet ($8; for manual cutting only)

https://www.amazon.com/dp/B09SFC8J56

- Final size – 77.5 mm x 40.3 mm, with 6.3 mm hole

Laser Cut Acrylic Cover Lens with Engraved Paw Print

Partial Kit of Parts

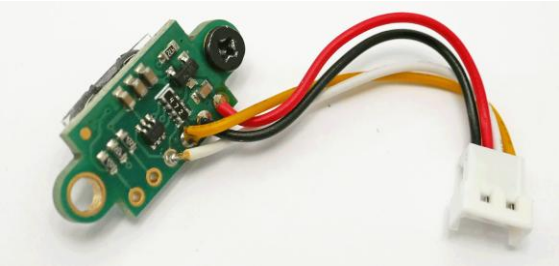
We made a kit of select parts to make the project easier and at a lower cost ($23). These kits are available on eBay while supplies last at: https://www.ebay.com/itm/267592151520

This kit includes 2 - M3 x 8 mm screws; 6 - M2.3 x 4mm screws; an injection molded 6mm blue button; 4 - 10mm diameter magnets; a male and female connector pair; and a laser cut lens cover made from smoke transparent acrylic with a paw engraving. You will still need to source/buy/make the other parts.

Building Pet Detector

Wiring

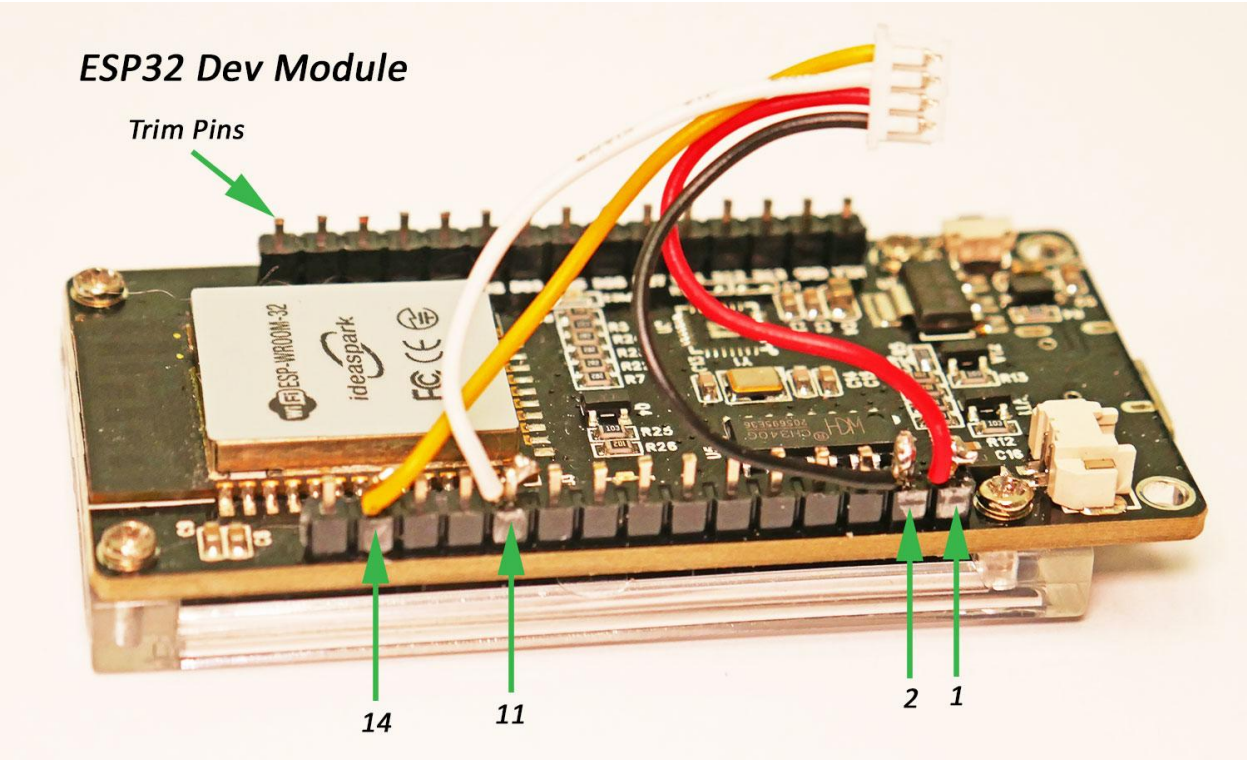
You can directly connect the ESP32 module to the TOF sensor (4 wires) or use the wired connectors as we did. If using the connectors, only use 4mm / 1.5” of wires from each connector.



On the ESP32 Module, trim the pins down to 2mm. Use diagonal pliers to cut them down. It would have been nice to buy these without the headers, but I never found such a version.

Solder wires from the connector to the processor board:

VL53L5X Labels	Color Wire	Ideaspark Pin	ESP32 Label
VCC	red	1 (3.3v)	3V3
GND	black	2	GND
SDA	white	11	D21 (IC2 data)
SCL	yellow	14	D22 (IC2 clock)



3D Printed Files

All of the 3D STL print files are located at:

https://github.com/frankvg/Pet-Detective/tree/main/3D%20STL%20Print%20Files

3D Printing Tips and Notes

We used a smooth build plate. It really improves the case front surface and the blackout frame parts, but it should work fine with a stock rough-surface plate too. The blackout frame should be black. The case and other parts can be any color you want. We tried both black and white, and preferred the white, as it matches our pet door color. We used PLA for all parts, but PTEG or ABS should work fine too. If the Pet Detector case or parts will be in the sun, then use PTEG or ABS.

When printing the case, support is needed. I recommend tree supports, and use the advanced support setting for object first layer gap as 1 mm. If you get any mushrooming on the first layers, you will need to adjust the Z axis setting. Typically, negative values push the hotend closer to the build plate (more mushrooming) and more positive value moves it upwards (less mushrooming). A small change can make a dramatic improvement.

Button

There is a single button for various functions. I first tried a 3D printed button and even at 0.08 mm layer height, the results were poor. I had some 6mm x 6mm button caps, that were injection molded. There was no flange on the button to keep it captured, so I did print a tiny button flange (0.2 mm layer height is fine), and epoxied it to the button.

One side of the flange is missing, so it fits without rubbing on the LCD display nearby.

On Amazon here is a kit of buttons and caps in different colors (far more than needed for this project, and you don't need the electrical buttons):
https://www.amazon.com/OCR-Tactile-Momentary-Switches-Colorful/dp/B07CMZCQS5
AliExpress also has many button choices at a much lower price, but may take a month to arrive. Seach for "button cap 6mm". Often these are listed as A56 buttons. A button is included in the parts kit on eBay.

Front LCD Cover Lens – Laser Cut Method

A laser offers the cleanest cut in acrylic, but not all lasers can cut transparent material. Do not use polycarbonate, as the fumes are toxic and it usually melts and makes a mess rather than cutting.

We created a cut dimensions to 77.5 mm x 40.3 mm with a 1mm corner radius in our laser software. We also added an engraved paw print. After cutting, we lightly filed the edges to clean off burs. We ended up not using the laser for the hole, as we were unable to find a setting that didn't melt it slightly around the hole. We drilled the hole instead (see the Fit and Drill section below)

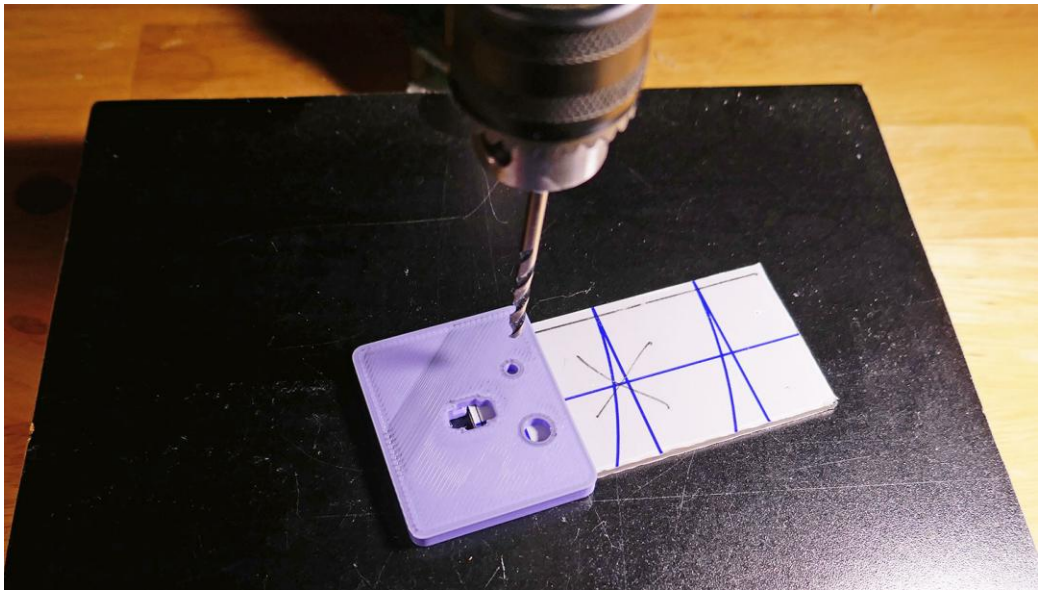
Front LCD Cover Lens – Manual Method

From sheet of acrylic or polycarbonate, cut a rectangular part 77.5 mm x 40.3 mm. Leave the protective film on until the very end to avoid scratches. There is plenty of material if you miscut it. Use the 3D template to help check the size as you cut. After cutting, We filed the four corners slightly, as the corners are sharp!

Front LCD Cover Lens – Fit and Drill

Test the fit in the 3D printed case front. It's intended to be a tight fit and a slight bit of flexing is necessary to snap it into place.

Remove the cover lens to add a 6.3 mm or 1/4" hole for the button. We created another 3D printed template to help position the hole. Position the template over the side of the cover. First drill the small pilot hole 3mm or 1/8".



Using a drill press for the first 1/16" hole

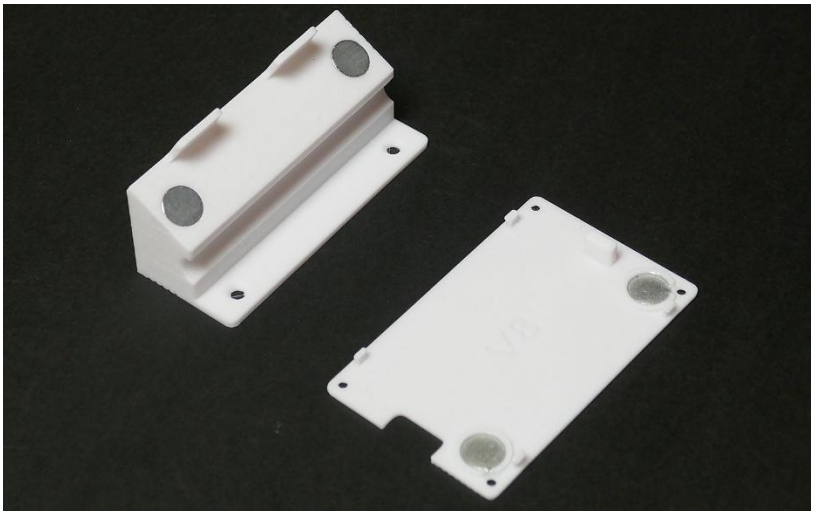
Flip the lens in the template and then drill the larger 6.3 mm / ¼" hole into the prior pilot hole. Ideally use a drill press and slowly lower the bit into the hole. If you lower too quickly, it can melt and/or catch and spin it around! Confirm the button fits in the hole without binding and enlarge with a round file if needed. Remove the film and clean any dust and debris with a soft cloth.

Magnets

The magnets are in a roll. To separate the magnets slide them apart. Lifting apart might break the magnet! Using a Sharpe pen, we recommend marking one side with a dot as you separate each magnet. This will help you keep the proper orientation with the "dot" side facing into the plastic. The magnet orientation between the parts is critical, as you need them to attract them rather than repel!

In the back cover insert the two magnets. Use disposable gloves to avoid getting epoxy on your hands and add some epoxy on the top to ensure they stay in place and are electrically isolated from the ESP32 module.

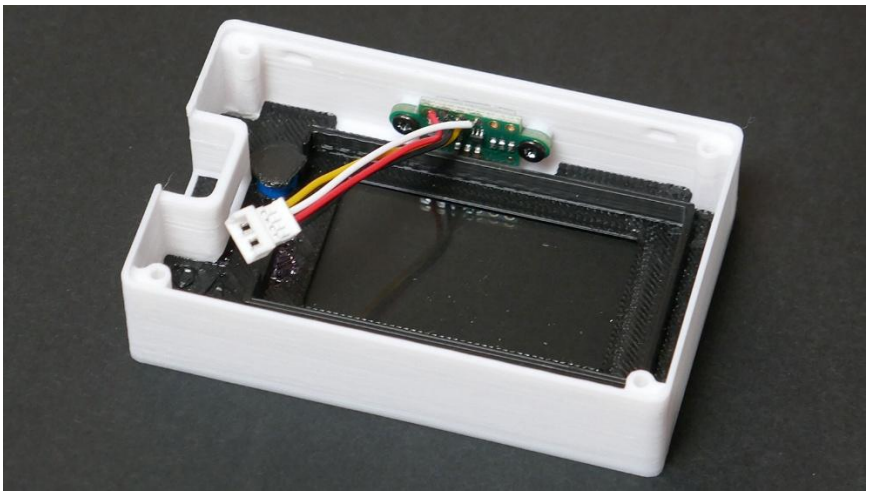
On the case angle mount, first check that the magnets dry fit. If they are too tight, clear away any filament overhang. Carefully put a small amount of epoxy in each of the two indentations. Press the magnets into the epoxy.



Magnets Installed on Mount and Rear Case

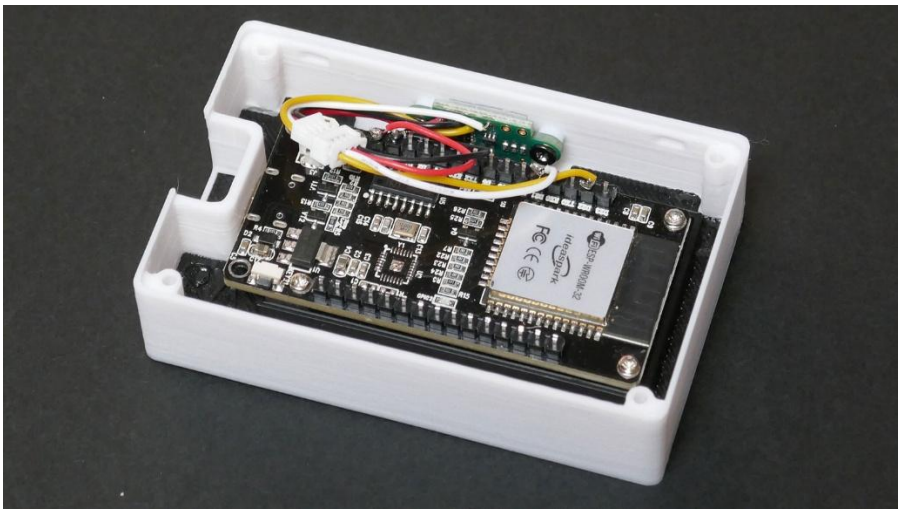
Assembly Order

- Insert the cover lens (some bending is required). The hole is oriented towards the two sides that have holes.
- Insert the blackout frame. It slides under the USB indent.
- Insert the button cap and confirm it moves smoothly.
- On the laser sensor, be sure to remove the clear protective film. Attached the dark red lens cap on the sensor that was included with the sensor purchase. Insert the sensor assembly into the bottom hole and use two M2.3 x 4mm screws to hold it in place.



The assembly with laser sensor installed

- Remove the protective film from the LCD. Slide the ESP32 module USB connector into the hole and press down so the blackout frame aligns it.
- Confirm the button cap smoothly presses on the ESP32 push button.
- Connect the laser sensor connector to the ESP32 connector (each previously soldered).



Sensor and Processor with LCD installed

- Attach the case rear. It should snap into place and is designed to hold the ESP32 in place. Insert the four M2.3 x 4 screws to secure the case.
- Install the 180° USB-C adapter (optional). This is not necessary but makes a cleaner look rather than the USB cable sticking out the side, especially if you want the cable to feed out the right side.

The Final Assembled Unit

Software Installation

You will need to download and install the Arduino IDE. Here are detailed instructions: https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE

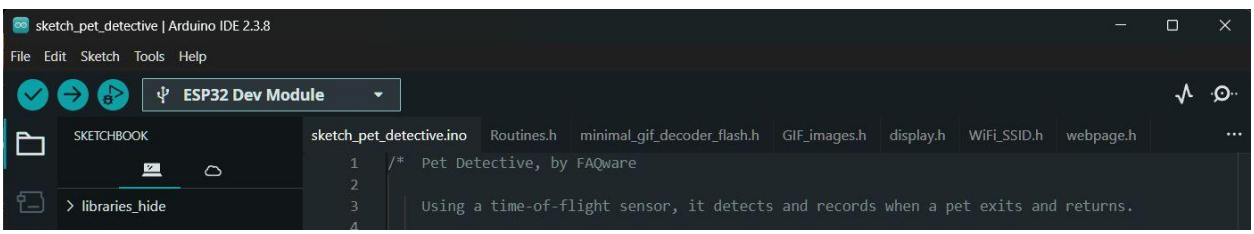
Next, install the Windows CH340 driver and set up the IDE to work with the ESP32. Follow steps 1 through 8 on this reference: https://manuals.plus/asin/B0D6QXC813. While the driver instructions suggest using Window 10, I've had no problems with Windows 11. We have no idea if this can be made to work with a Mac but suspect it can.

Download the Pet Detective source code. It is available on GitHub: https://github.com/frankvg/Pet-Detective

Copy all the files into a directory like: **Pet Detective**

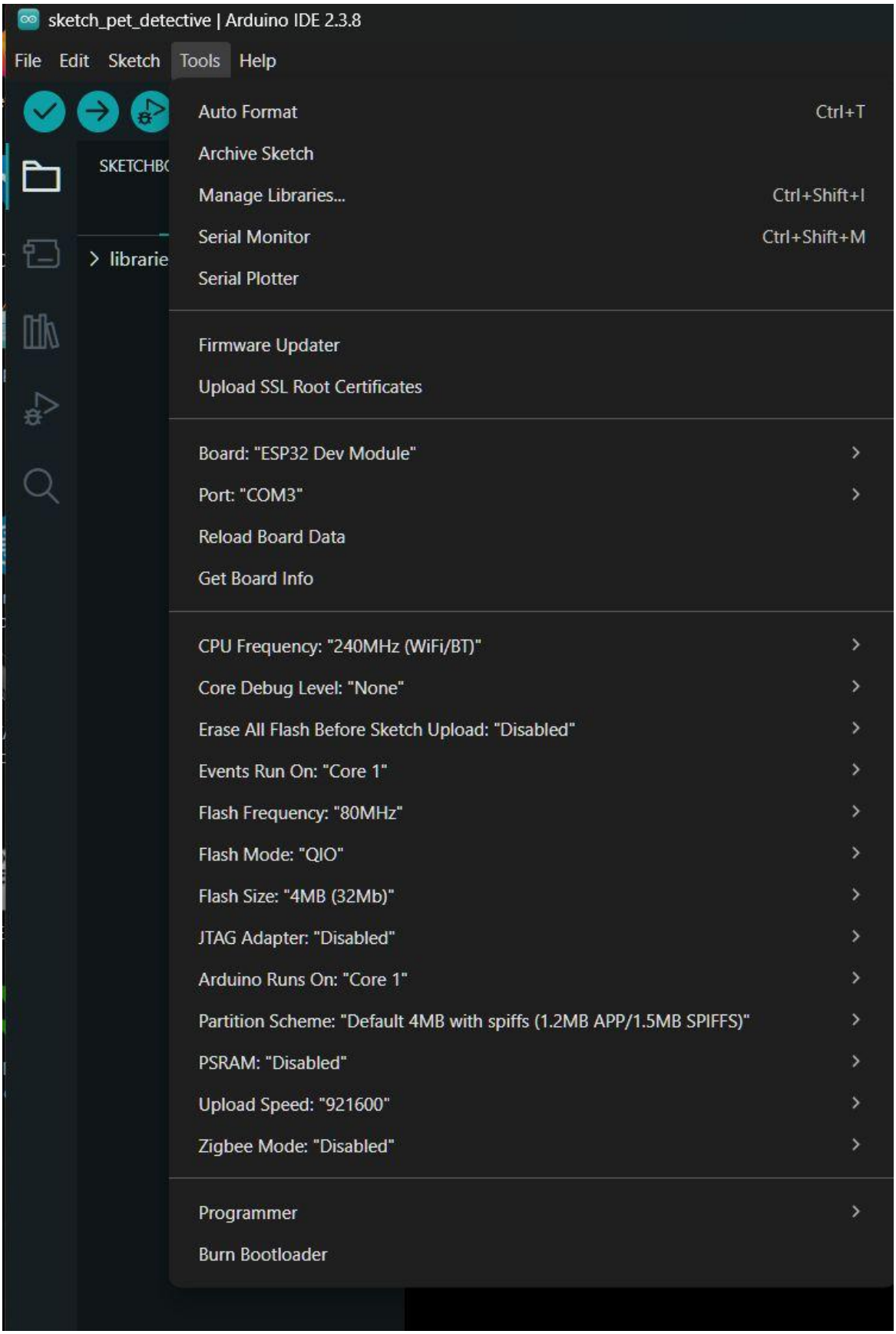
In the **sketch_pet_detective** directory, double-click on the file: **sketch_pet_detective.ino**

The IDE should open with the needed files.

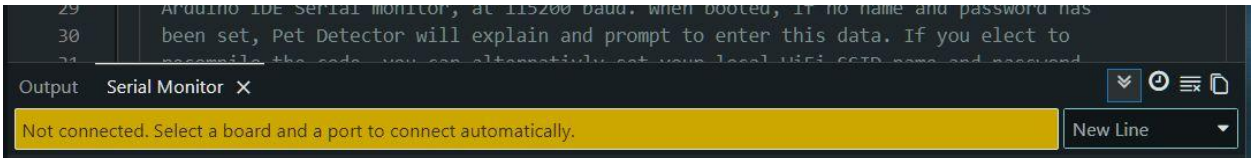


Connect the ESP32 module via USB to your PC. Check and/or set the COM3 connection in the IDE from the menu: **Tools -> Port**

With things set up properly, the Tools menu looks like this:

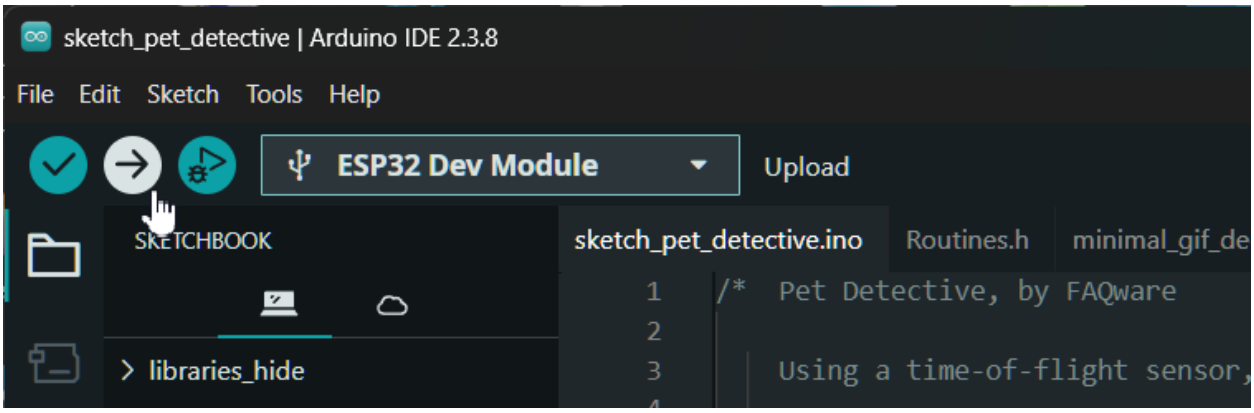

sketch_pet_detective | Arduino IDE 2.3.8
File
Edit
Sketch
Tools
Help
SKETCHBO
librarie
Auto Format Ctrl+T
Archive Sketch
Manage Libraries... Ctrl+Shift+I
Serial Monitor Ctrl+Shift+M
Serial Plotter
Firmware Updater
Upload SSL Root Certificates
Board: "ESP32 Dev Module"
Port: "COM3"
Reload Board Data
Get Board Info
CPU Frequency: "240MHz (WiFi/BT)"
Core Debug Level: "None"
Erase All Flash Before Sketch Upload: "Disabled"
Events Run On: "Core 1"
Flash Frequency: "80MHz"
Flash Mode: "QIO"
Flash Size: "4MB (32Mb)"
JTAG Adapter: "Disabled"
Arduino Runs On: "Core 1"
Partition Scheme: "Default 4MB with spiffs (1.2MB APP/1.5MB SPIFFS)"
PSRAM: "Disabled"
Upload Speed: "921600"
Zigbee Mode: "Disabled"
Programmer
Burn Bootloader

On the Tools menu, select **Serial Monitor**. If you get the amber error:



Something is not set up right and you'll need to resolve it. One common error is using a USB cable that is power only (no data). The Serial monitor should be set to 115200 baud.

Select the download button (see the hand icon below) to compile and install the software into the ESP32 module.



When properly connected and booting for the first time, in the Serial monitor you'll see it ask for the WiFi SSID and password:

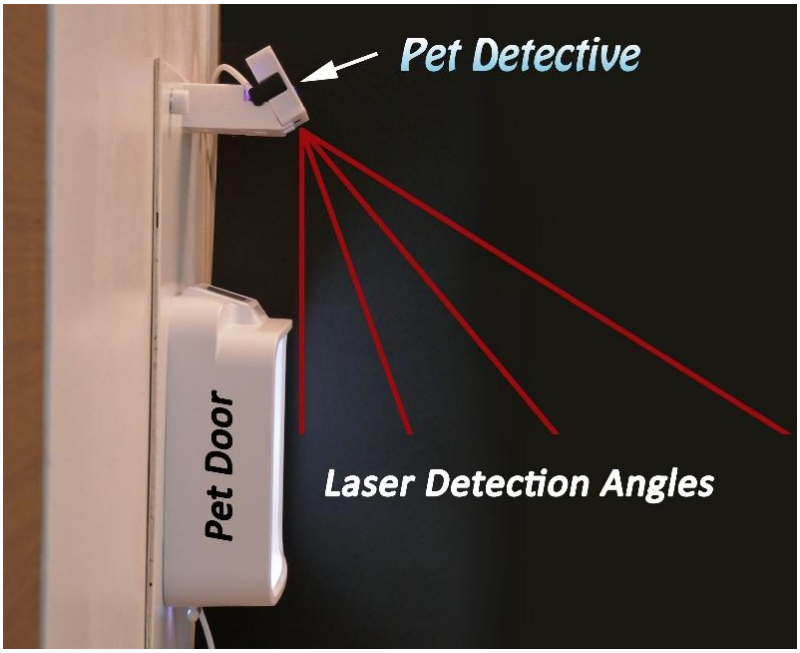
Enter your WiFi information when requested or press Enter twice if you don't plan to use WiFi. Once set, the data is remembered and will not ask again. If you need to enter or change it later, when Pet Detective is installed and running, use short button presses to get

to the **About** menu, and long press (10-seconds) to do a factory reset. With a reset, the WiFi data is cleared, and the IDE Serial Monitor will prompt again for the SSID and password. As an alternative, you can also hard code the SSID and password in the WiFi_SSID.h file.

Mounting Pet Detective

The Pet Detective case must be installed above the pet door at a 30-degree angle, so the laser detection covers the area in front of the pet door. The case angle mount provides the 30-degree angle needed. This also makes the display easier to see.

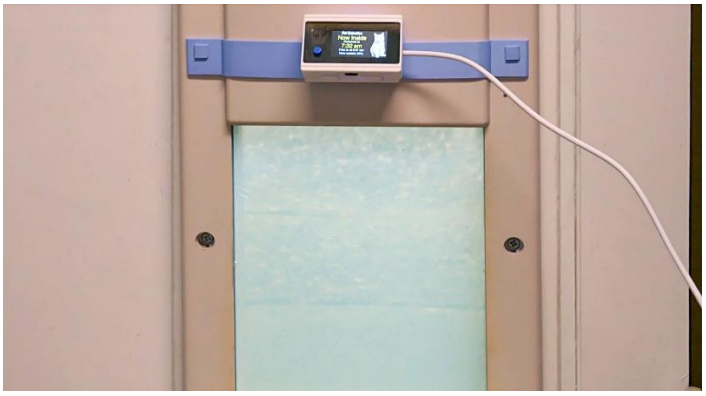
The mount must extend out so that the first downward facing laser detection beam is in front of the door. This image shows a side view with the combination of the extended mount, the case angle mount and the Pet Detective case.



Side View Laser Angles with Extended Mount

Other pet doors may not need the extended mount and/or may need a different mounting. If you do need a different extension size, it is easy in your slicer to change the Z size to something different.

If you have the medium size High Tech pet door, we made a bracket that holds the Pet Detective on existing holes in the Pet Door. The design includes two screw hole covers. We happen to print these in blue PLA.



Using Pet Detector

To test Pet Detector, you only need to move your hand forward or backwards under the sensor. You will see color bars on the left showing when each laser detector triggers. After each detected transition, we have a 12 second transition lockout. We found this avoided duplicate transitions when a pet moves slowly.

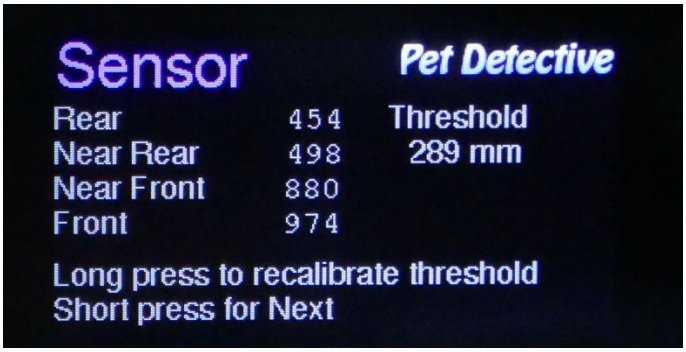
Menus

A short press of the button will rotate through the various menus. While in menu mode, it will not detect pet transitions. If left in a menu, after 5 minutes of no activity, it will revert to normal monitoring mode. Settings, other than time and log entries, are saved in non-volatile memory.

Log – This shows the last 4 transitions. You can rotate through the 8 pages of the log with a long (1 second) press. It shows the date and time of each transition, along with trend data. Every 100ms, we count trends of laser threshold detection in each direction to help determine a valid transition.

WiFi – Shows the SSID you set, the MAC address, the web address to remotely monitor status and log data, and the signal level. A long press in this menu is used to reset WiFi and reacquire the time.

Sensor – Shows the distance detected in millimeters for each of the four laser detectors. It also shows the current sensor threshold. Use a long press to recapture a new threshold value. Threshold is set at 60% of the shortest distance measured on the four laser detectors. If you are getting erratic transition results, changing the threshold often makes a dramatic improvement.



Once a threshold is set (either from this menu or the very first time powered up), it is retained in non-volatile memory. All later power-up sequences will use the saved threshold value.

Hours – Use a long press to toggle between 12-hour and 24-hour time.

Time Zone – Use a long press to switch to the next time zone and DST (Daylight Savings) mode. The time will be updated immediately on the display.

Color – There are 3 colors in 6 combinations. Each long press toggles through the 6 combinations. A very long 10-second press toggles the brightness. The default is maximum brightness, intended for the dark smoke acrylic lens. If using a clear or light polycarbonate lens, the dimmer setting may be preferred.

About – Basic information. A very long press (10-seconds) resets all values to factory defaults. A short press ends menu mode and returns to normal transition monitoring.

Transition Accuracy

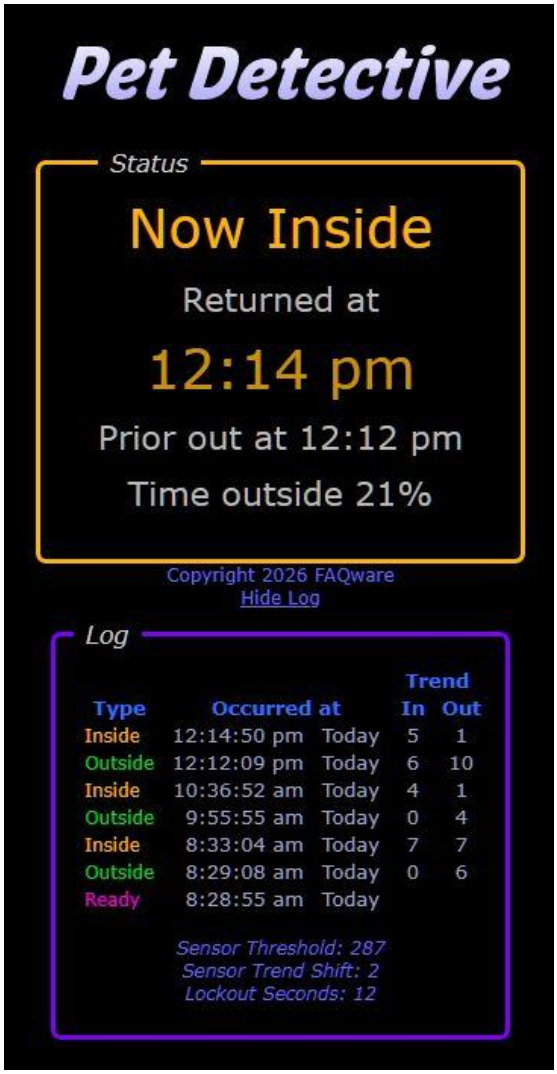
If your pet leaves or returns when not going through the pet door, Pet Detective will not be aware of the change. We sometimes let our pet in through a nearby door. If wanted, you can sweep your hand under the sensor to set a fake “Now Inside” status.

We’ve also had a few rare occasions when the pet is inside, he comes up to the door, looks out, but doesn’t exit. Depending on the angle and how they move, it could trigger a gone outside transition when he is still inside.

Browser

Using the URL from the WiFi menu, you can view the status and the full log in any browser on the same local network as Pet Detective. Note that you must use a http:// prefix, and

not https://. Most new browsers will automatically switch to https://, which will then give you an “Unable to connect” error. If this occurs, do a Google search for “option to allow http://” for your browser and follow the instructions. Heres’ an example with the status and log view enabled:



Status changes are checked every 6 seconds. Web page updates are ignored when you are in a menu.

Troubleshooting

Button - Failure to reliably detect button presses can occur if the processor is a bit overloaded. Updating the web page on your local network, going out to the internet to get the time, or transitioning the display can each take far more than the allocated 100ms loop time, causing a short button press to be ignored.

Fails to detect pet transition – In almost all cases the sensor threshold is set too short. When booted it shows the saved threshold. You can also view the threshold in the Sensor menu. Reset the threshold from the Sensor menu. A good value is the distance from the sensor to a depth about ½ below the top of the pet's body.

Duplicate pet transitions – These are tricky to fix if you have the sensor threshold set properly. Be aware if your pet comes up to the door, waits, and then leaves without making a transition through the door, it may trigger a duplication. This can also occur if your pet is either let out or in through a different door, so a transition state is missing.

Look at the sensor trend data to see if a different sensor trend shift might help. This can only be changed in the source code and really shouldn't be necessary.

Notes for Software Developers

Pet Detector software consists mostly of Arduino C++ for the ESP32 processor. It also includes some HTML for the web page, and a small amount of JavaScript to only perform web page updates when a transition occurs.

Graphic Images

Graphics for the four pet photos are small GIF images, that are saved as byte data in program memory. If you wish to replace a GIF image, you will need to make the non-animated GIF (the same size as we use), 256 colors, and use a black background. Turn off transparency. Convert the image into an array of bytes for C++ and replace the existing array for that graphic. Here is the online GIF to File array converter we use: https://mischianti.org/online-converter-file-to-cpp-gzip-byte-array-3/

The code will decode the GIF content into a displayable format for the LCD (RGB565 pixels). The cat image byte arrays are stored in GIF_images.h

Startup Pet:	141 x 140	Cat_Detective_Jacket_141x140.gif
Ready Pet:	89 x 170	Cat_Unknown_89x170.gif
Inside Pet:	93 x 170	Cat_Inside_Sitting_93x170.gif
Outside Pet:	123 x 170	Cat_Outside_Sitting_123x170.gif

We realize this is a bit complex to change an image, but it is necessary as the ESP32 module has no SD Card or file system to store a GIF file. The ESP module is also a bit slow, so we suspect if an SD card and file system were included, it would be far slower than having the data already in program memory. We did test pre-loading the images in RAM but didn't notice a significant difference in display performance. Generally displaying one of the four pets above takes about 1 second to display. A one-second update is not an issue, as the display changes infrequently.

Sensor

We use a Time-of-Flight laser sensor that can detect the distance of 64 different locations over a 60x60 degree spread. We only use the faster 4x4 mode (16 locations), and then only look at 4 of these locations. When the module is powered on for the first time, it does an automatic calibration, assuming it is installed and has no pet in sight. This is saved and used from that point onwards. The software limits the distance threshold to between 300 mm and 80 mm. Even if you were blocking the sensor during calibration, it's likely close to what works anyway.

The sensor menu allows you to view the real-time sensor distance measured in millimeters. From this menu, you can also reset the sensor threshold. Once the threshold is set, either at the very first-time boot, or manually from the sensor menu, it is saved and does not change in future power/boot cycles.

A "secret" trace menu is available on the Sensor menu by holding the button for 10 seconds. It shows all the data collected for a single transition. Every 100ms, each of the four triggered sensors are shown in a log of values in a single screen. When one full transition cycle is completed, it shows the Trend values and if it is a Inside or Outside transition, based on the Trend shift value. We used it for tuning the design.

WiFi

Pet Detector uses WiFi to obtain and maintain the current time. It will work fine without WiFi, but it will not be able to time-stamp events. To use WiFi, during connection to the IDE, during the first powerup, you will be prompted to enter your WiFi name (SSID) and password. Optionally, if you plan to recompile, you can hard code WiFi info in the WiF_SSID.h file.

The ESP32 module has a built-in antenna on the left side. It does require a good signal, so if your WiFi router is too far away and/or has too many walls between it and the Pet Detector, you may not be able to make WiFi work.

Security

This does so little, there is no need for web site encryption. Should someone hack your local WiFi network, they might see if your pet is inside or outside If they happen to connect during the few seconds when a transition occurs and you are looking at the web page.

There are no remote downloads into the ESP32 nor other ways they could hack Pet Detective. Web encryption (https://) would slow web page display considerably, as the ESP32 processor is not intended to be a powerful server. As it is, the simple Pet Detective server sending a full web page, with log, consumes about 2 seconds of processing time. This doesn't cause a problem as the web page is only updated during a pet transition during the 12-second lockout period.

Pet Detective goes to the internet to get the date and time. There is no code in Pet Detective to go to any other server or external connection.

We default using the most common time reference: pool.ntp.org. You are welcome to change it to any other standard NTP server in the code. Should the time server fail to provide the time, Pet Detective tries again later. Once the time is acquired, it checks periodically to accurately maintain the time. If you have a power failure, it automatically acquires the time upon power up.

Future Enhancements

Sound
We are considering adding an amplifier and small speaker to create a sound for incoming and outgoing events. The ESP32 module does not have an output that can drive a speaker directly, so an extra amplifier is needed. Since there is no file system, it also means the WAV file needs to be converted into a large byte array. There may not be enough remaining code space to include such a feature.

Phone/Tablet Status
We may create an Android app that displays a widget of the current Pet Detective status. This would only work when the phone or tablet is connected to the local network. Always more to do!

© 2026 FAQware